

The proposed system is designed as a Conversational Recommender System (CRS) that integrates large language models (LLMs), structured data processing, and retrieval-augmented

generation (RAG) to provide personalized vehicle recommendations through natural language interaction. The whole process is described below as well as on the figure 1.

Data Preprocessing

The first data was extracted from the website CarAPI [8]. It returns data in JSON format, which was then processed into SQLite table (on 1 marked as Structured database).

Process: We sent the parsed query to that table which returned a list of multiple car candidates. constraint-based filtering.

Another data in pdf format was extracted from the website Auto-Brochures [9]. The website consist of several different brands of cars, found on the US market. We focused on 26 of them. With Qwen embedding model we created embedded vectors from that dataset. (on 1 marked as FAISS Vector database). Mention FAISS

Process: we embed the parsed query and send it to that dataset

CRS System

kako deluje etc

LLM prompts etc

how we created prompts: ur assistant task rules (no json, just natrual language)

how memory works (it remembers all queries and last LLM prompt)

The method begins with a natural language interface through which users describe their preferences and requirements. These inputs are processed by an LLM to extract structured representations of user intent, such as budget constraints, vehicle type, and qualitative preferences.

The system then employs a dual data strategy. A structured database containing quantitative vehicle attributes (e.g., price, fuel consumption, engine specifications) is used to perform constraint-based filtering and reduce the candidate set. In parallel, a vector database built from unstructured sources from brochures and catalogs enables semantic retrieval of relevant contextual information. We will use Auto Catalog Archive [10] as our initial dataset. This website contains more than 55,000 car brochures that include many different car models. Each brochure provides technical specifications (like power, weight, range...), list of features (like type of wheel, roof rail, safety equipment...), and other promotional information.

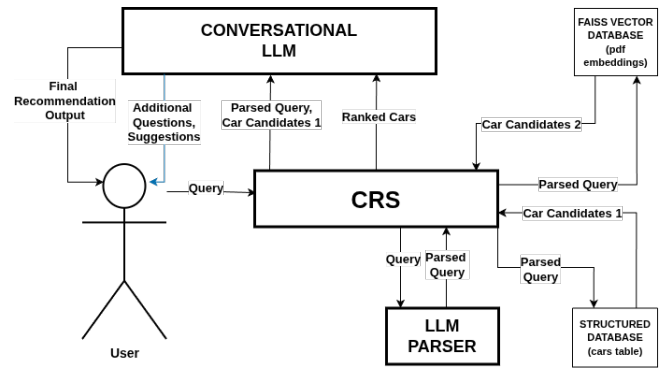


Figure 1. CRS System in a Diagram

Evaluation of a model

Evaluation ideas:

Test if RAG is working well

Build a small controlled dataset (10 very different cars), where only one is meeting the requirements. Ask the system to recommend a car and check if it picks only the valid one.

Compare CRS to other LLMs

Another idea is also creating a few prompts, for which we find a set of suitable cars. Then we compare the output of our CRS system and one another LLM (preferably one that doesn't have access to web). Then we evaluate if the LLMs found any car from the set.

Compare CRS to some website

Compare recommendation to some website (Carwow, Auto-trader according to chatgpt) and compare for the same requirements the models output, though it is subjective method of evaluation (hard to tell which model is better, no hard truth on this topic)

Advanced method - baseline filter model

Ideally (but too complicated to implement in such short time) would be to create a basic filtering system, and compare the outputs with what the LLM is offering. Again the idea wouldn't be to show it is better output objectively but either it shows more option, or wilder choices, etc. Especially if the requirements are not precise or not coherent between themselves (for instance big car + low consumption + high acceleration)

Equations

Lists

Figures

Tables

Code examples

Results

Discussion

Use the Discussion section to objectively evaluate your work, do not just put praise on everything you did, be critical and exposes flaws and weaknesses of your solution. You can also explain what you would do differently if you would be able to start again and what upgrades could be done on the project in the future.

Acknowledgments

Here you can thank other persons (advisors, colleagues ...) that contributed to the successful completion of your project.

References

- [1] Amira Riham Beddouda and Bochra Belhemissi. *A VEHICLE RECOMMENDATION SYSTEM FOR SALES*. PhD thesis, KASDI MERBAH UNIVERSITY OUARGLA.
- [2] Saumya Singh, Soumyadepta Das, Ananya Sajwan, Ishanika Singh, and Ashish Alok. Car recommendation system using customer reviews. *International Research Journal of Engineering and Technology (IRJET)*, 9(10):983–989, 2022.
- [3] Lizi Liao, Ryuichi Takanobu, Yunshan Ma, Xun Yang, Minlie Huang, and Tat-Seng Chua. Deep conversational recommender in travel, 2019.
- [4] ZK Abdurahman Baizal, Dede Tarwidi, B Wijaya, et al. Tourism destination recommendation using ontology-based conversational recommender system. *International Journal of Computing and Digital Systems*, 10, 2021.
- [5] Juan Carlos Carrillo, Victoria Beltran, Laura Sebastia, and Eva Onaindia. Smarttur+ eco: a conversational recommender system for tourism. 2023.
- [6] Yueming Sun and Yi Zhang. Conversational recommender system. In *The 41st international acm sigir conference on research & development in information retrieval*, pages 235–244, 2018.
- [7] Yaochen Zhu, Chao Wan, Harald Steck, Dawen Liang, Yesu Feng, Nathan Kallus, and Jundong Li. Collaborative retrieval for large language model-based conversational recommender systems. In *Proceedings of the ACM on Web Conference 2025*, pages 3323–3334, 2025.
- [8] Carapi, 2026.
- [9] Auto-brochures.
- [10] Auto catalog archive, 2016.